Identifying Signals of Unmet Need Using NHS System Data

A reproducible framework for specialty-level screening and prioritisation

NHS England

The challenge: Research and innovation investment decisions are often informed by policy priorities and clinical expertise. However, there is limited use of routinely collected NHS data to systematically identify areas of unmet need.

This project: Developed a transparent, data-driven framework to explore whether NHS system indicators can provide an additional evidence layer to support prioritisation.

Explore and harmonise data → **Generate and test signals in Python** → **Visualise national and regional patterns in Power BI**

Proof-of-concept scope:

42	4	3	4	2
INTEGRATED CARE BOARDS (ICBs)	SPECIALTIES	DOMAINS	OPERATIONAL INDICATORS	CONTEXTUAL INDICATORS

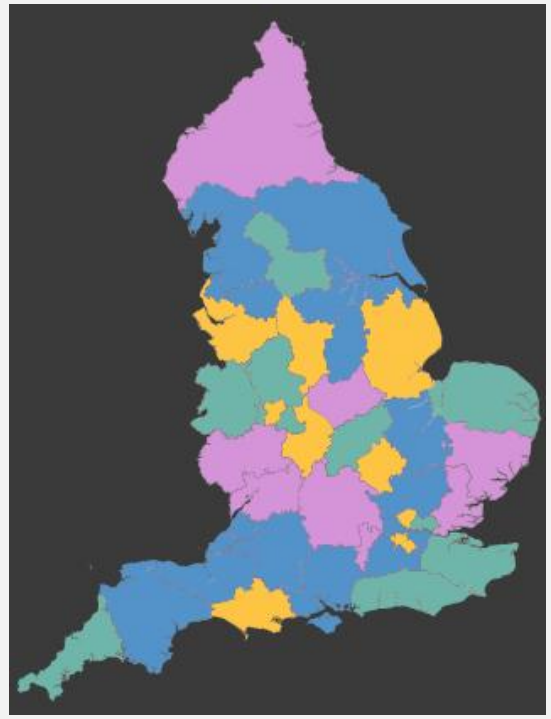
Figure 1: National specialty overview



Figure 2: Regional view: leading signal by ICB

This highlights how the leading specialty signal varies between ICBs.

Colour = specialty with the highest composite signal in each ICB



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